



OPEN Pneumonia detection from enhanced chest X-Ray images based on Double SGAN model

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Medical imaging plays a crucial role in clinical diagnosis, however, deep learning models often struggle with imbalanced datasets, which has a negative impact on the accuracy and robustness of pneumonia image classification. This study proposes a deep learning based diagnostic system for pneumonia detection using chest X-ray images. Using the pneumonia MNIST dataset, including pediatric lung images. To address the issue of class imbalance and improve generalization ability, we innovatively propose a Double SGAN model. Firstly, apply spectral normalization to all generator and discriminator layers for stable training and improve performance. Secondly, self-attention mechanisms are integrated into the convolutional layers of the generator to better capture complex image features. Finally, the hinge loss function is used during the training process to further improve learning efficiency. On this basis, this study constructed the ResNet18-SA classification model, which embeds spatial attention mechanism in the residual module of the lightweight network ResNet18 to focus on key feature regions related to pneumonia diagnosis and suppress background noise interference. The experimental results show that the ResNet18-SA model outperforms traditional models in terms of accuracy, precision, recall, and F1 score, reaching 95.83%, 95.87%, 95.21%, and 95.52%, respectively.

Keywords Deep learning, Spectral normalization, Chest X-ray, Image classification, Generative adversarial network.

Pneumonia is a respiratory disease caused by viruses, fungi, or bacteria, early and accurate diagnosis is crucial for vulnerable populations, such as the elderly and children. Chest X-rays are the preferred imaging modality, as they can quickly reveal the extent and severity of lung infections, providing critical evidence for timely treatment. Due to their low cost and ease of operation, chest X-ray has become the primary imaging method for pneumonia screening and diagnosis. However, traditional manual radiograph reading is highly subjective and inefficient, often leading to misdiagnosis and missed diagnoses, especially in primary care settings. Therefore, the development of deep learning-based pneumonia X-ray image classification methods¹ has become a research focus in recent years in the field of medical image analysis.

With the development of convolutional neural network technology, numerous studies have confirmed the effectiveness of deep learning in pneumonia X-ray image classification. Khan et al.² improved the classification accuracy of COVID-19, pulmonary opacity, normal, and pneumonia images to 96.13% by fine-tuning the EfficientNetB1 network and introducing regularization techniques in the classification head, effectively verifying the advantages of deep transfer learning in COVID-19 X-ray image classification. Ibrahim et al.³ proposed a multi classification method for chest X-ray images based on pretrained AlexNet network. The experimental results showed that the method achieved the highest accuracy of 93.4% in the corresponding task, confirming that chest X-ray combined with deep learning can be used as a fast, low-cost, and accurate computer-aided screening tool.

In addition to chest X-rays, intelligent auscultation technology based on respiratory sounds (RS) has also become a new low-cost, radiation free solution for lung disease screening in recent years. Roy et al.⁴ proposed the Pulmo-TS2ONN model, which achieved a recognition accuracy of 98.88% for seven lung diseases on three publicly available respiratory sound datasets by fusing triple time-frequency features. Abera Tessema et al.⁵ independently developed an electronic stethoscope that uses discrete wavelet denoising and multi-resolution feature extraction techniques, combined with Bayesian optimized Fine Gaussian Support Vector Machine (Fine Gaussian SVM), to classify seven types of lung diseases, including pneumonia. These studies indicate that respiratory sounds can provide high-quality diagnostic clues without relying on ionizing radiation and expensive equipment, providing a new path for supplementing X-ray deep learning diagnostic methods in resource limited

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environments. In addition, Boumans et al.⁶ conducted a systematic evaluation and analysis of lung ultrasound (LUS) for the first time, with 2075 adult patients included in 11 studies. Due to the characteristics of non radiation, repeatable operation at the bedside, and suitability for low resource environments, LUS can complement deep learning frameworks based on chest X-rays, and is easy to fuse with clinical features or respiratory sound features to achieve multi label joint detection, further enriching the technical system of pulmonary disease diagnosis.

Although the above methods continue to break records in classification accuracy, their performance is highly dependent on large-scale, high-quality, and class balanced training data. In practical applications, medical imaging data often faces serious class imbalance problems due to the difficulty of annotation and high computational costs. This not only leads to poor performance of the model on minority class samples, but also limits the model's generalization ability. To alleviate this problem, researchers have attempted to use data augmentation techniques⁷ to obtain sufficient samples. Currently, commonly used methods include geometric transformations (such as rotation, scaling, cropping), noise addition (such as Gaussian noise, salt and pepper noise), and signal translation (such as horizontal or vertical translation). However, these traditional methods can only increase the sample size to a certain extent and are essentially simple variations of the original samples, unable to introduce new semantic information or generate more discriminative features for the dataset.

In recent years, image generation techniques based on Generative Adversarial Networks⁸ and diffusion models⁹ have provided new ideas for enhancing medical imaging data. Heng et al.¹⁰ proposed a novel medical image synthesis method called HLSNC-GAN based on the CycleGAN framework, aiming to generate computed tomography (CT) images from magnetic resonance imaging (MRI) data. The framework incorporates switchable normalization techniques that improve the quality of synthesized images while avoiding mode collapse and gradient problems. Chen et al.¹¹ conducted research on the generation of chest X-ray images using deep convolutional generative adversarial networks (DCGANs) and found that the slope loss of the training loss regression line can reflect the stability of training and the quality of generated images. By analyzing the slope of the training loss regression line, they optimized the GAN training process and improved the efficiency and effectiveness of medical image generation.

Sharma and Guleria et al.¹² conducted a systematic literature review to explore deep learning methods for detecting pneumonia based on chest X-ray images. They delved into the application of convolutional neural networks (CNNs), pretrained models, and ensemble models in pneumonia detection, evaluated the effectiveness of these models in addressing challenges in the medical field, and described in detail the model architecture and workflow. They conducted in-depth analysis on performance indicators, hyperparameter tuning, and model fine-tuning. Huijben et al.¹³ first systematically validated the data augmentation potential of the conditional denoising diffusion probability model (conditional DDPM) in multi label classification of chest X-rays. By synthesizing a sample balanced training set, the sensitivity of multi label classification of pneumonia was significantly improved from 0.02 to 0.59, confirming the practical value of the diffusion model in solving data scarcity and class imbalance problems. Cahan et al.¹⁴ proposed a cross modal transformation framework based on diffusion models, which can generate high contrast three-dimensional computed tomography pulmonary angiography (CTPA) images from low-cost two-dimensional chest X-rays, improving the specificity of pulmonary embolism diagnosis. Miyato et al.¹⁵ analyzed the 1-Lipschitz condition of convolutional networks and concluded that a convolutional network with all layers satisfying the 1-Lipschitz condition also satisfies the condition as a whole. They proposed a new weight normalization technique to stabilize the GAN training process. In the scenario of imbalanced dataset categories, Drir et al.¹⁶ proposed a novel DCGAN model (DCGAN With VGG16) that integrates perceptual loss¹⁷ and VGG16 network, effectively solving the problems of data scarcity and category imbalance in photovoltaic model fault classification, and improving the accuracy of complex defect classification. Zhou et al.¹⁸ proposed a small ACGAN model, which expands the dataset and improves the accuracy of plant disease recognition by improving the network architecture and loss function. Aljawarneh et al.¹⁹ confirmed that using an improved CNN model and transfer learning model²⁰ for data augmentation can achieve an accuracy of 92.46% for the enhanced model. Ranjan et al.²¹ propose a method that combines autoencoder enhanced GANs with CNNs and Zero Sample Learning to address the issue of limited labeled data in hyperspectral image classification.

In conclusion, although significant progress has been made in the classification methods of chest X-ray images²², data scarcity and class imbalance²³ remain critical bottlenecks that constrain model performance. Traditional augmentation methods are nearing saturation, while image generation-based augmentation strategies are gradually becoming the frontier of research due to their high quality, controllability, and diversity. Therefore, addressing the core pain points of medical imaging, namely "low signal-to-noise ratio, sparse lesions and imbalanced categories", this paper innovatively proposes a data augmentation method based on image generation. By improving the deep learning model Double SGAN through collaborative design tailored to medical scenarios, it aims to accurately increase the minority class samples in the training set, thereby significantly enhancing the accuracy and robustness of pneumonia detection tasks. The contributions of this paper can be summarized in four key points:

1. In this study, we initially integrated spectral normalization techniques into the generator and discriminator to improve the training stability and generation performance of the model when processing chest X-ray images. By limiting the maximum singular value of the weight matrix, this technique ensures the Lipschitz continuity of the network, effectively alleviating the problems of gradient explosion and vanishing. Meanwhile, a self attention mechanism was introduced in the generator to enable the model to better focus on important features in the image.
2. Introduces a hinge loss function tailored specifically for generative adversarial networks. By minimizing the difference between the outputs of the generator and discriminator, it enhances the stability of network training and ultimately greatly improves the quality of the generated X-ray images.

- 3. This study constructed the ResNet18-SA classification model, which embeds spatial attention mechanism in the residual module of the lightweight network ResNet18. By performing average and max pooling, convolution operation, and sigmoid activation on the feature maps, spatial weight coefficients are generated to focus the model on key feature regions related to pneumonia diagnosis, suppress background noise interference, and further optimize feature extraction efficiency.
- 4. Validated using the MNIST pneumonia dataset, the experimental results demonstrate that the method proposed in this study exhibits superior performance in image classification tasks of pneumonia. Compared to the EfficientNetV2L model, our method achieves an improvement of 1.81% in accuracy and 1.47% in precision, offering an effective deep learning approach for the field of medical image analysis.

Related work
The basic principles of GAN

In recent years, the field of medical imaging has further explored a new hybrid deep learning paradigm of “deep learning+automatic report generation”. Gupta et al.²⁴ proposed the UDCT framework, which combines U-Net with Differential Architecture Search (DARTS) to achieve end-to-end detection and segmentation of lung cancer lesions on CT images. Its automated network design strategy provides structural optimization ideas for subsequent model construction; Gupta and Kumar²⁵ proposed a lightweight hybrid model with parallel CNN+Transformer branches, which significantly improved the accuracy of lung disease classification and verified the robustness of the hybrid architecture in sample imbalance scenarios; Sharma et al.²⁶ proposed the FDT-Dr2T model, which uses DenseNet to encode visual features and combines Transformer decoder to automatically generate X-ray radiation reports, providing end-to-end templates for interpretable diagnosis.

It is worth noting that another type of exploration in this field focuses on optimizing data efficiency and model adaptability, and Amin’s team’s series of studies is quite representative: The deep active learning framework proposed by Amin²⁷ in 2023 enhanced classifier performance by 2.1% for chest X-ray classification tasks while utilizing only 65% of labeled data, thereby effectively reducing the annotation cost of medical data. However, this method exhibits a strong dependence on the quality of original images. Earlier, Amin²⁸ developed the EADN model, which adopts a lightweight 3D-CNN architecture integrated with a sparse attention mechanism to achieve video anomaly detection. The “lightweight backbone + attention” design paradigm of this model provides critical technical reference for the lightweight transformation of medical imaging models, particularly in reducing model parameters and improving inference efficiency to adapt to clinical deployment scenarios. In 2024, Amin et al.²⁹ further advanced this research by integrating the EfficientNet-B0 backbone with the CBAM mechanism. This integration achieved a 4.7% improvement in mean Average Precision (mAP) for epidemic monitoring video tasks, validating the effectiveness of dual-attention fusion under a lightweight backbone. Nevertheless, this study has two notable limitations: first, it has not been migrated to medical scenarios; second, the compatibility of its spectral normalization technique and the stability of gradient propagation have not been verified in medical model frameworks. In addition, Amin³⁰ realized multi-task classification of COVID-19, tuberculosis, and pneumonia by fusing five independent CNN models trained on chest X-ray data, achieving an Area Under the Curve (AUC) of 0.906. Despite this promising performance, the work only focuses on “multi-model integration at the discriminative end”, optimizing classification results through ensemble learning and does not address the optimization of quality and diversity for synthetic data at the generative end, a key gap given the scarcity and class imbalance of medical imaging datasets. In order to visually demonstrate the current research limitations, we summarize them in Table 1.

Although the above research has made breakthroughs in segmentation, classification and report generation, its performance is still limited by high-quality, category balanced training data. Therefore, this paper turns to the deep generative models, they are a class of powerful unsupervised machine learning models³¹ that have played a significant role in various applications such as image generation, compression learning and domain adaptation. Generative adversarial networks, a model consisting of a generator and a discriminator, are one of the important approaches in the field of generative models. The task of the generator is to learn the distribution of real data and generate synthetic samples that are indistinguishable from real images; the objective of the discriminator is to distinguish between real samples and generated samples in order to maximize the discriminative accuracy. The description of the principle of the GAN algorithm is described as shown in Equation 1:

min_G max_D V(D, G) = E_{x ~ P_{data}(x)} [ln D(x)] + E_{z ~ P_z(z)} [ln(1 - D(G(z)))] (1)

Ref.	Model	Limitations
24	UDCT	Lack of exploration on interpretability of model decisions
25	Parallel CNN+Transformer	Insufficient data diversity
26	FDTD-X-DT	High model complexity and slow inference speed
27	Deep Active Learning Framework	Strong dependence on the quality of original images
28	EADN	No mention of model lightweighting and hardware adaptability
29	EfficientNet-B0 + CBAM	Compatibility of its spectral normalization technique not verified
30	Fusion of 5 Independent CNN Models	Ignores generative-end synthetic data diversity

Table 1. Comparison table of related research in medical imaging field based on GAN.

In the equation, G and D represent the generator and the discriminator respectively, $V(D, G)$ indicating the difference value between G and D , $\mathbb{E}_{x \sim P_{\text{data}}(x)}$ denotes the mathematical expectation, x represents real samples, z denotes random noise, $G(z)$ represents generated samples, and $D(G(z))$ represents the probability of the discriminator judging the input sample to be the real sample, $P_{\text{data}}(x)$ and $P_z(z)$ denote the distributions of the true sample and the generated sample respectively.

Traditional GANs struggle to learn complex data distributions due to their relatively shallow network architectures, which can lead to difficulties and instabilities during training. To address these issues, DCGAN³² eliminates traditional fully connected layers and extensively employs convolutional and deconvolutional layers to enhance the performance of image processing. Additionally, the incorporation of batch normalization (BN)³³ and Leaky ReLU activation functions³⁴ in both the generator and discriminator significantly improves the quality of generated images while increasing the stability of the training process.

Residual model

Currently, there are many mature neural network models, with the earliest being LeNet³⁵. Subsequently, several classic neural network models have been proposed, such as AlexNet³⁶, VGGNet³⁷, GoogleNet³⁸, and ResNet³⁹, among others. This study mainly focuses on the optimization and improvement of the ResNet model. This network primarily introduces the residual learning module, whose core idea is to perform feature aggregation of shallow network features and those learned in deep networks through identity mapping, and to continue learning with this aggregated input in the next layer. In traditional neural networks, the input to each layer undergoes a series of linear transformations and nonlinear activation functions before being output to the next layer. Due to inevitable signal loss during the layer-by-layer transmission process, issues such as gradient vanishing and explosion arise as the network depth increases, which limits the depth of the network. The resnet architecture addresses this issue by incorporating residual structures. The residual module has a simple structure, primarily consisting of a residual learning branch and an identity mapping branch for the output. By summing the two branches and passing them through a nonlinear activation function, a complete residual module is formed. Stacking the aforementioned residual modules allows for an increase in network depth while maintaining the same performance as a shallower network, thereby addressing the problem of network degradation caused by increased depth. The formula for the residual module is shown in Equation 2.

$$y = \phi(F(x, W) + x) \quad (2)$$

where y denotes the output of the residual block, ϕ is the activation function, $F(x, W)$ represents the residual function, x is the input feature, and W stands for the weight parameters. Right now, there are ResNet18, ResNet34, ResNet50, ResNet101, and ResNet152 in residual network models.

The attention mechanism

The attention mechanism has emerged as a key technology in deep learning, widely applied in the fields of natural language processing and machine vision. In image processing tasks, the attention mechanism can simulate the way human vision focuses, granting higher attention to key areas of information within images, thereby enhancing the model's discriminative ability in complex scenarios. In recent years, the generation of countermeasures network is no longer limited to image to image conversion, but gradually expanded to medical image enhancement, severe weather removal and other more detailed tasks. Bose et al.⁴⁰ proposed a multi spatial attention condition GAN framework for backlight images: the generator uses U-Net and embeds spatial attention in each encoder to synchronously complete feature extraction and down sampling; The discriminator uses a 1×1 PatchGAN structure, combined with user-defined loss of the L1+MSE+SSIM composite, to significantly improve detail recovery and color fidelity of the backlight area. Kolekar et al.⁴¹ focused on the rain stripe removal task, whose generator is a stacked form of "residual U-Net+spatial attention". The discriminator uses multi-scale attention to guide CNN. Through the weighted combination of adversarial, L1 compliance and VGG perception loss, it achieves the current optimal PSNR and SSIM on Rain100H and other datasets. It is worth noting that these two works have verified the effectiveness of "spatial attention mechanism + lightweight cGAN" in small samples and high dynamic scenarios.

This article accordingly implements a differentiated design within the structure. By incorporating self-attention mechanisms in the generator, the aim is to enable the network to establish long-range dependencies among global pixels when processing chest X-ray images. This article employs the ResNet18-SA network to weight the spatial dimensions of the feature maps. This allows the model to focus more on the key areas that are most relevant to pneumonia diagnosis, suppressing background noise interference and enhancing the model's sensitivity to lesion features, thereby improving classification accuracy and robustness.

Methods

Figure 1 presents the overall framework of the approach proposed in this paper, which can be summarized into two main modules: image processing and model training. The image processing module initially reads the raw chest X-ray images. Due to class imbalance in the dataset, this paper inputs the data into the proposed Double SGAN for data augmentation and balancing. The generated images are merged with the real samples after quality screening, forming a balanced dataset with equal class numbers for subsequent training and testing. The model training module then feeds the processed images in batches into the ResNet18-SA network, monitoring the loss in real time during the training process; upon convergence, inference is performed solely on the original test set, calculating accuracy, precision, recall, and F1 score, and comparing these with classical CNNs while plotting a confusion matrix to comprehensively validate the effectiveness of the proposed Double SGAN.

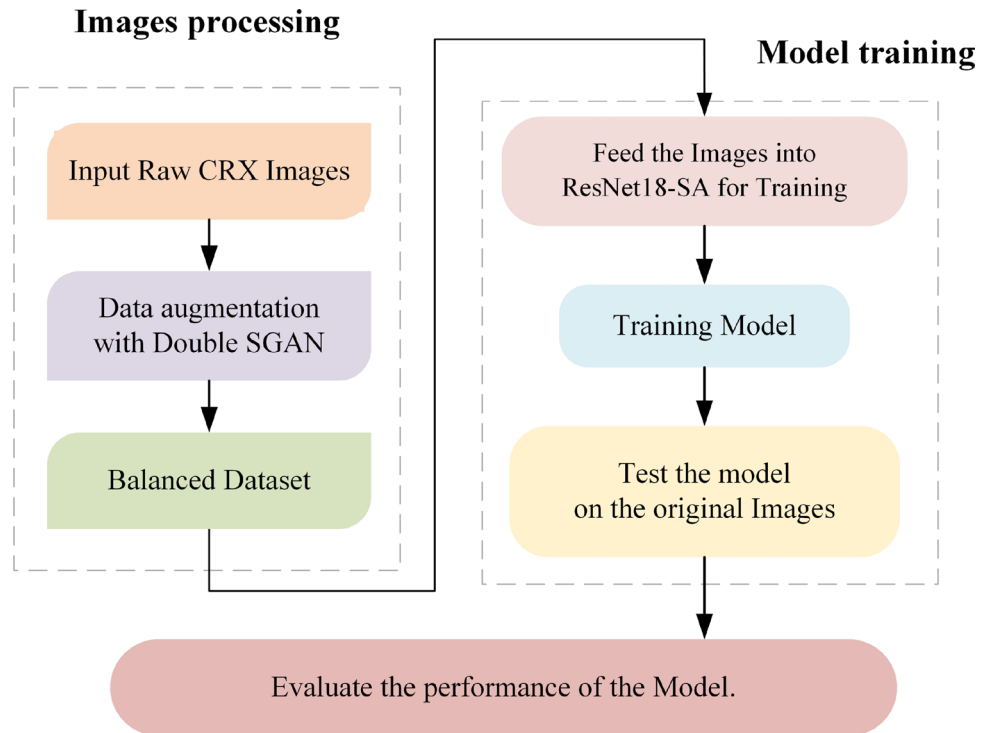


Fig. 1. Architecture of Chest X-ray Image Classification System Based on Double SGAN and ResNet18-SA. First, data augmentation is performed on the original chest X-ray images using Double SGAN to generate a balanced dataset. Second, the balanced samples are fed into the ResNet18-SA network for training; finally, the model's performance is evaluated on the original test images.

Double SGAN model

This article proposes an improved dataset augmentation model, Double SGAN, which is systematically expanded and optimized based on the architecture of deep convolutional generative adversarial networks, as shown in Fig. 2. The explanation of Double is mainly reflected in the following three aspects. Firstly, all convolution kernels of the generator and discriminator undergo spectral normalization simultaneously. Its function is to ensure bidirectional convergence of 1-Lipschitz; Secondly, this means that this article uses two types of attention mechanisms. On the one hand, integrating self attention mechanisms into the generator to improve the quality of synthesized images; On the other hand, introducing spatial attention into the residual classification network allows the model to focus more on the features of the focal region; Thirdly, this means that the generator and discriminator use hinge loss during training to provide continuous large gradients and prevent gradient saturation.

In the generator of the Double SGAN model, as shown on the left side of Fig. 3, a 100-dimensional random noise vector, denoted as Noise z ($z=100$), following a normal distribution is initially input into the generative network (G). This noise vector acts as a seed in the latent space for synthesizing new image samples. The generator is composed of four layers of transposed convolutional layers, each incorporating spectral normalization¹⁵ techniques, while layer normalization⁴² is applied after each layer, replacing the batch normalization traditionally used in DCGAN. Layer Normalization (LN) normalizes the feature dimensions of a single sample rather than relying on batch statistics; therefore, it is insensitive to batch size and demonstrates greater stability in scenarios with small batches or variable-length inputs. Moreover, LN can mitigate internal covariate shift, enhancing the model's generalization ability when there is a distribution discrepancy between the training and test sets. After the transposed convolution layer, a ReLU activation function⁴³ is used. Compared to Tanh or Sigmoid activation functions, ReLU is non-saturating, which effectively alleviates the vanishing gradient problem and accelerates the convergence of deep networks. Its linear characteristics also help the generator to learn more continuous mapping relationships in the latent space, thereby effectively enhancing the diversity of generated samples. At the end of the generator, an attention mechanism module is added to enhance the spatial consistency and detail expressiveness of the images. This module computes the global correlations between feature maps, enabling the generator to focus on critical areas within the image and establish pixel relationships across distant images.

During the discrimination phase, as shown on the right side of Fig. 3 the generated images are input into the discriminator together with real images from the original dataset, ultimately yielding a realism probability score. The output of the discriminator is a real number between 0 and 1, indicating the probability that the output image is a real sample; the closer the value is to 1, the more realistic the image is considered, while a result of 0 indicates a fake image. The discriminator is composed of several convolutional layers, and spectral normalization is also applied to constrain its Lipschitz constant, thereby further reducing the risk of oscillation.

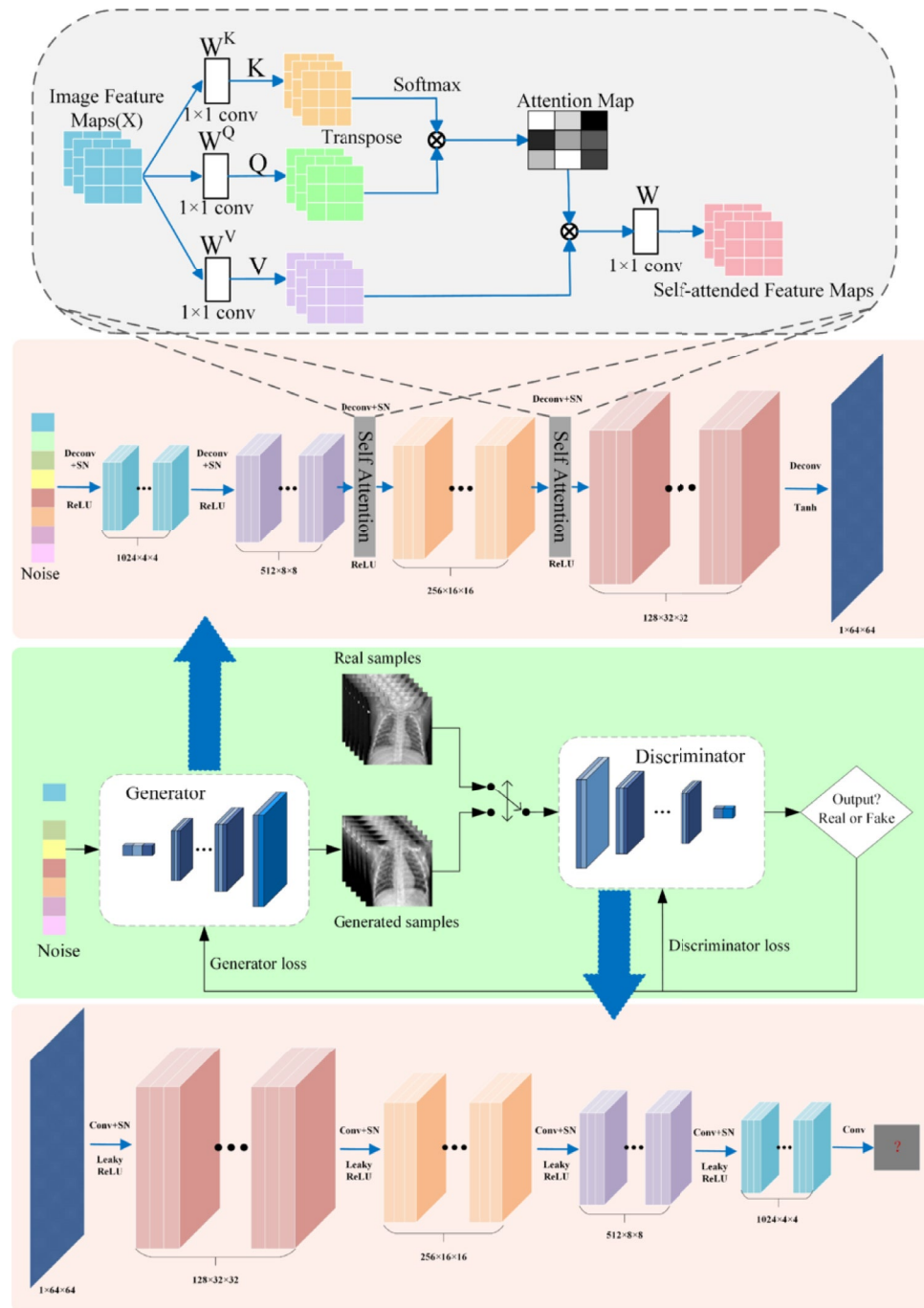


Fig. 2. Overall architecture diagram of Double SGAN pneumonia data augmentation network.

in the model during the training process. In order to prevent the discriminator from becoming overly powerful, we introduced a gradient penalty term⁴⁴ into the discriminator, as shown in Equation 3, allowing the generator and discriminator to maintain a dynamic balance. Through this series of improvements, we effectively enhanced the training efficiency and stability of the model.

$$\mathcal{L}_{GP} = \mathbb{E}_{\hat{x} \sim \mathcal{P}_{\hat{x}}} \left[(\|\nabla_{\hat{x}} D(\hat{x})\|_2 - 1)^2 \right] \quad (3)$$

It punishes the degree to which the L2 norm of the sample gradient deviates from 1, prompting the model to satisfy the Lipschitz continuity condition and improve training stability.

Spectral normalization

In this paper, we focus particularly on innovative improvements in the application of regularization techniques⁴⁵. Specifically, this paper not only introduces SN in the discriminator but also comprehensively applies this

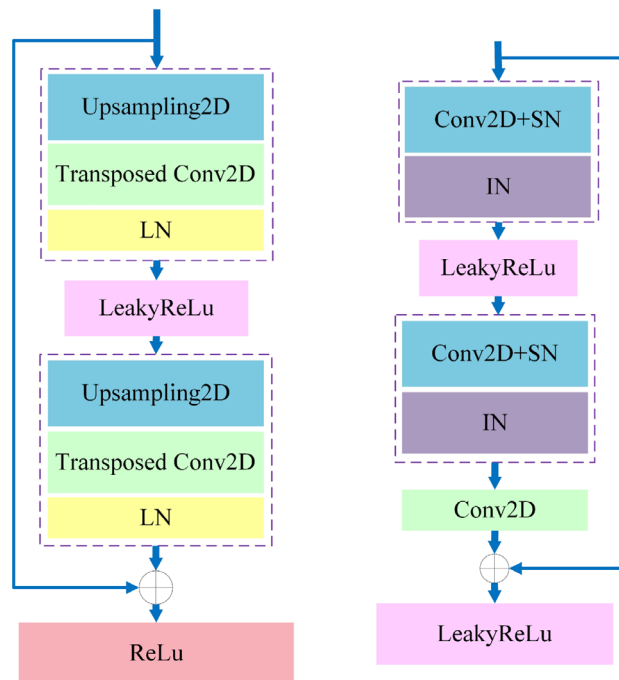


Fig. 3. Detailed network structure diagram of the generator (left) and discriminator (right) in Double SGAN.

technique to every layer of the generator network as well. The core mechanism of spectral normalization, as an efficient means of regularization, is to constrain the Lipschitz constant⁴⁶ of the network by limiting the spectral paradigm of the weight matrix. The application of this technique not only significantly enhances the stability of the discriminator but also improves the training efficiency of the entire DCGAN system as well as the quality of the generated images.

In the traditional training process for Generative Adversarial Networks (GAN), maintaining a balance between the discriminator and the generator is essential. An overly powerful discriminator can inhibit the generator's learning process, negatively affecting the quality and diversity of the generated images. By applying spectral normalization constraints to both the generator and the discriminator, one can effectively regulate the discriminative and generative capabilities of both components. This balanced training approach helps prevent scenarios in which an excessively dominant discriminator stifles the generator's learning, thereby enhancing both efficiency and stability during training. Moreover, a significant advantage of employing spectral normalization technique is that it eliminates the need for additional hyper-parameter tuning. In practical applications, this characteristic greatly simplifies the optimization process of the model, reducing the complexity of model training. Specifically, the implementation steps of spectral normalization technique are as follows: Firstly, a Singular Value Decomposition (SVD)⁴⁷ is performed on each weight matrix to obtain its singular values along with their corresponding left and right singular vectors. Subsequently, the spectral normalization coefficients are calculated, with the maximum singular value serving as the spectral normalization coefficient for each weight matrix, which is defined as shown in Formula 4.

$$\sigma(D) = \max_{x \neq 0} \frac{\|Dx\|_2}{\|x\|_2} = \max_{\|x\|_2 \leq 1} \|Dx\|_2 \quad (4)$$

In the equation, x is a non-zero vector ($x \neq 0$), it represents an element input to the matrix D . The coefficient σ represents the magnitude of the largest eigenvalue of the weight matrix, which $\sigma(D)$ is the spectral norm of the matrix D . For each layer of the network g , there is a specific formulation $\|g\|_{Lip} = \sup_x \sigma(\nabla g(x))$. For a general linear layer, $g(x) = Wx$, there is $\|g\|_{Lip} = \sigma(W)$, the parameters W are denoted as a parameter matrix. If g denotes the output layer of a multi-layer network, and W^n denotes the parameter matrix of the n -th layer, Equation 5 can be expressed as:

$$\|g\|_{Lip} \leq \prod_{n=1}^{N+1} \sigma(W^n) \quad (5)$$

The matrix is normalized by dividing it by the spectral normalization coefficient, thereby constraining the Lipschitz constant of the network to be less than 1, as shown in Equation 6.

$$W_{SN} = \frac{W}{\sigma(W)} \quad (6)$$

This method prevents the weight matrices in each layer of the network from becoming excessively large, thereby controlling the Lipschitz constant of the network function, which facilitates the regularization of the network.

Hinge loss

Traditional CNNs widely employ binary cross-entropy (BCE) loss in classification or generation tasks; however, its limitations are particularly evident in complex scenarios such as medical imaging. BCE loss is extremely sensitive to class imbalance, leading to minority class samples being easily overwhelmed by the majority class. When the predicted probabilities approach 0 or 1, gradients quickly saturate, causing the generator to stall in training due to a lack of effective signals. Extreme misclassifications are excessively penalized, forcing the model to accommodate noise at the expense of generalization. Moreover, BCE lacks a margin maximization mechanism, being content with merely correct judgment rather than confident judgment, resulting in narrow decision boundaries and poor robustness. Under the GAN framework, this can lead to a situation where a strong discriminator causes gradients to approach zero, triggering mode collapse and generating blurry images. To overcome these shortcomings, this paper introduces Hinge loss¹⁰, which restricts the discriminator outputs within the ± 1 range through a hard margin constraint, preventing gradient disappearance while consistently providing large gradient signals. This allows the generator to steadily approach the true distribution in a stable adversarial setting, thereby enhancing the quality and diversity of the synthesized chest X-rays.

Hinge loss effectively reduces the difference between generated images and real images by optimizing the adversarial process between the generator and discriminator, thereby improving the quality of synthesized images. Specifically, the Hinge loss penalizes the difference between the images generated by the generator and the real images, incentivizing the generator to output results that are closer to real images. In generative adversarial networks (GANs), the generator is dedicated to producing images that are indistinguishable from real images, while the discriminator focuses on identifying the authenticity of the images. The Hinge loss function plays a crucial role in this adversarial mechanism by constraining the range of gradients, enhancing the stability of the GAN, which in turn helps the model converge more smoothly during the training process, effectively alleviating the problem of gradient explosion, and thus achieving a more stable and effective training process. For the discriminator, the Hinge loss function can be defined as shown in Equation 7.

$$L_D = \max(0, 1 - D(x)) + \max(0, 1 + D(G(z))) \quad (7)$$

Where $D(x)$ is the output of the discriminator for the real image x , $G(z)$ is the image generated by the generator based on random noise z , and $D(G(z))$ is the output of the discriminator for generated images. This loss function encourages the discriminator to output values close to 1 for real images and values close to -1 for generated images. For the generator, the Hinge loss function can be defined as shown in Equation 8.

$$L_G = -\min(0, -1 + D(G(z))) \quad (8)$$

This loss function encourages the generator to produce images that yield outputs from the discriminator close to 1, thereby generating more realistic images. In this way, the Hinge loss helps to reduce the disparity between generated images and real images, thus enhancing the quality of the generated images.

ResNet18-SA

This study mainly optimizes and improves based on the ResNet18 model. The ResNet18 residual neural network model is shown in Fig. 4, it shows the overall process of ResNet18 from a macro perspective. ResNet18 consists of four residual modules, each containing two convolutional layers, with a total of eighteen trainable layers. The input image first passes through the 7×7 convolution (stride=2, padding=3), then after the maximum pooling (3×3 , stride=2), and finally passes through four residual blocks in turn. The number of channels is doubled group by group from $64 \rightarrow 128 \rightarrow 256 \rightarrow 512$. There are two 3×3 convolutions stacked in each group and the short circuit connection is reserved. The final feature map space size is reduced to 1/32 of the input, and the channel dimension is expanded to 512. So as to be used for feature mapping and category judgment in subsequent classification tasks.

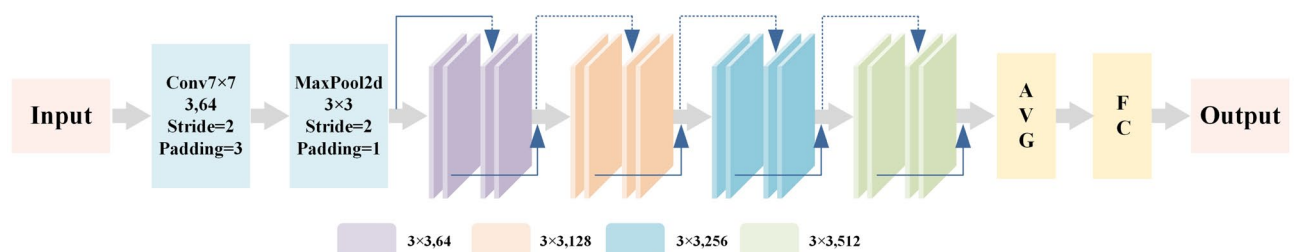


Fig. 4. Overall architecture diagram of ResNet18 residual neural network.

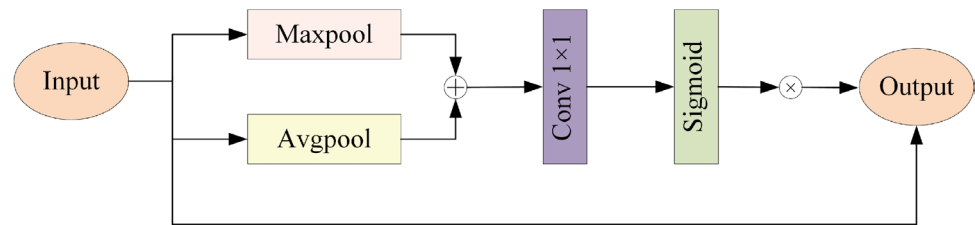


Fig. 5. The spatial attention module structure in ResNet18-SA.

(a) Parameter settings of Double SGAN model		
Parameter	Meaning	Value
H×W	Size of the input image	64 × 64
Epoch	Number of training epochs	400
Batch_size	The number of images per training batch	128
Output_H×W	Dimensions of the generated image	64 × 64
Lr_D	Discriminator learning rate	0.0002
Lr_G	Generator learning rate	0.0002

(b) Parameter settings of ResNet18-SA model		
Parameter category	Specific settings	Supplementary information
Optimizer	Adam	Effectively balance training stability and convergence speed
Epoch	50	Number of training epochs
Batch_size	64	The number of images per training batch
Initial Learning Rate	0.0001	Adjust according to the dataset complexity
Early Stopping	20	Prevent overfitting and improve generalization

Table 2. Model parameter setting table in experimental environment (including Double SGAN and ResNet18-SA).

ResNet18, as a lightweight version of the deep residual neural network, has advantages such as a small number of parameters, fast training speed, and strong generalization capabilities, and is therefore widely used in computer vision tasks such as image classification and object detection. Although ResNet18 performs well in feature extraction, there are still some issues when processing chest X-rays, such as insufficient utilization of global contextual information, primarily due to the limited receptive fields of traditional convolution operations, which makes it difficult to establish long-range dependencies. Additionally, there is inadequate focus on key areas; the lesion areas in chest X-ray images are often small and low in contrast, leading them to be overshadowed by background information. To address these issues, this paper adds an independent spatial attention(SA) branch to the feature layer after convolution operation in each residual block of the original ResNet18.

Its input is a feature map with dimensions $H \times W \times C$. Initially, average pooling and max pooling operations are performed on the input data separately, resulting in two channel feature matrices with dimensions $H \times W \times 1$, which are then concatenated. Subsequently, the convolution kernel and the Sigmoid function are utilized to obtain the spatial weight coefficients. Finally, the new weight coefficients are multiplied by the original feature map to obtain the new feature map. The structure of the spatial attention module is illustrated in Fig. 5.

After generating high-quality and diverse chest X-ray images using Double SGAN, ResNet18-SA processes the enhanced images and dynamically focuses on key areas using SA, which increases the model's sensitivity to lesion characteristics and helps avoid being misled by irrelevant information present in the generated images, thereby improving both the efficiency and accuracy of pneumonia detection.

Experiments

Experiment environment and hyper-parameters

The experiment was conducted on the Ubuntu operating system, utilizing an RTX 3080 Ti GPU. The development language used is Python 3.6, with Pytorch as the development platform and Pycharm as the development tool.

In this section, we will provide a detailed introduction to the key parameters involved in the experimental process. These parameters are crucial for the performance of the model as they directly affect the learning process and the quality of the generated images. The selection of parameters usually needs to be adjusted according to specific tasks and datasets to obtain the best training results. Table 2 shows the parameter settings used in this experiment, where Table 2a shows the training parameters of the Double SAGN model and Table 2b shows the training parameters of the ResNet18-SA model. Specifically, this dataset consists of 64×64 low-resolution images with low sample feature dimensions, which means that a sufficiently high learning rate is not necessary to fully update the parameters. If the learning rate is greater than 0.0005, it is prone to cause a gradient explosion in the generator. If the learning rate is less than 0.0001, the gradient may be too weak, leading to convergence

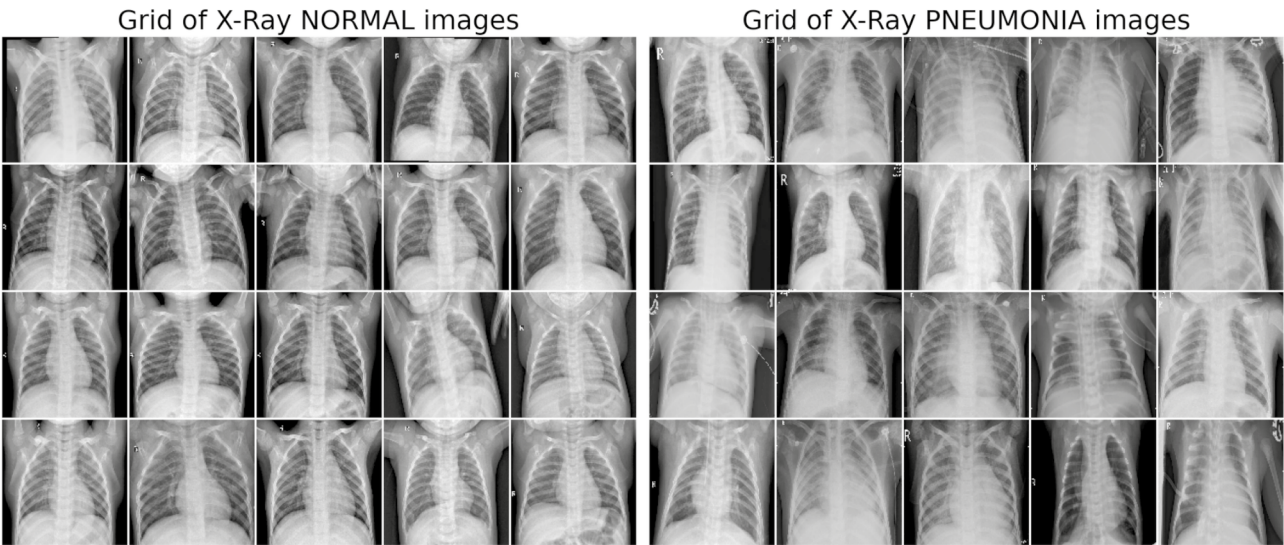


Fig. 6. Comparison of normal and pneumonia chest X-ray images in the original dataset.

Dataset	Numbers	Normal	Pneumonia	Ratio
Train	4708	1214	3494	1: 2.878
Val	524	135	389	1: 2.881
Test	624	234	390	1: 1.667
Total	5856	1583	4273	1: 2.699

Table 3. Sample distribution table of Pneumonia MNIST original dataset.

stagnation. To maintain the adversarial balance, the learning rates of the generator and the discriminator need to be consistent to avoid one side becoming overly dominant. Combined with the constraint effect of bidirectional spectral normalization on the gradient, a learning rate of 0.0002 can enable both to gradually approach a Nash equilibrium, ensuring that the discriminator can effectively distinguish between samples while providing sufficient update momentum for the generator.

Datasets and preprocessing

In this study, we utilized the pneumonia X-ray dataset from the MEDMNIST⁴⁸ project, which aggregates numerous medical image datasets, encompassing a variety of medical imaging technologies such as X-rays, OCT, and ultrasound. Due to its standardized processing workflow and diversified task types, this dataset holds immense value in the field of medical image analysis research. This article provides an in-depth analysis of the Pneumonia MNIST dataset⁴⁹, which is specifically designed for pneumonia image classification. It is based on chest X-ray images of 5,856 patients, with the core task being to determine whether inflammation is present in the lungs, that is, to differentiate between healthy and pneumonia-affected lungs, constituting a typical binary classification problem⁵⁰. To intuitively visualize the types of data in the images, Fig. 6 presents a comparison of normal and pneumonia X-rays. The grid on the left shows an X-ray image of a healthy chest, while the grid on the right displays the X-ray image of a chest affected by pneumonia.

This paper divides the dataset into three folders: training set, validation set, and testing set, where each folder contains sub-folders for two image categories: normal and pneumonia. The number of images in the subfolders is shown in Table 3. It is evident that the data in the training set is imbalanced, which tends to lead to a significant decline in the classifier performance. Therefore, this paper considers using Double SGAN to generate chest X-ray images to enhance the original dataset. We use peak signal-to-noise ratio ($PSNR \geq 30dB$) and structural similarity index ($SSIM \geq 0.80$) to evaluate the clarity and structural integrity of synthesized images. At the same time, these images were visually displayed, Fig. 7 illustrating the sample images generated by the model. This paper only considers the data of the normal category in the augmented training set. During the testing phase, only the original real images will be used to evaluate the performance of the model. The size of the augmented dataset is shown in Table 4.

Evaluation metrics

Currently, the assessment of the quality of images generated by Generative Adversarial Networks (GANs) is primarily divided into qualitative evaluation⁵¹ and quantitative evaluation⁵² methods. Qualitative evaluation relies mainly on human visual perception to judge the authenticity of generated images. However, due to significant perceptual differences among individuals, this method exhibits a high degree of subjectivity and

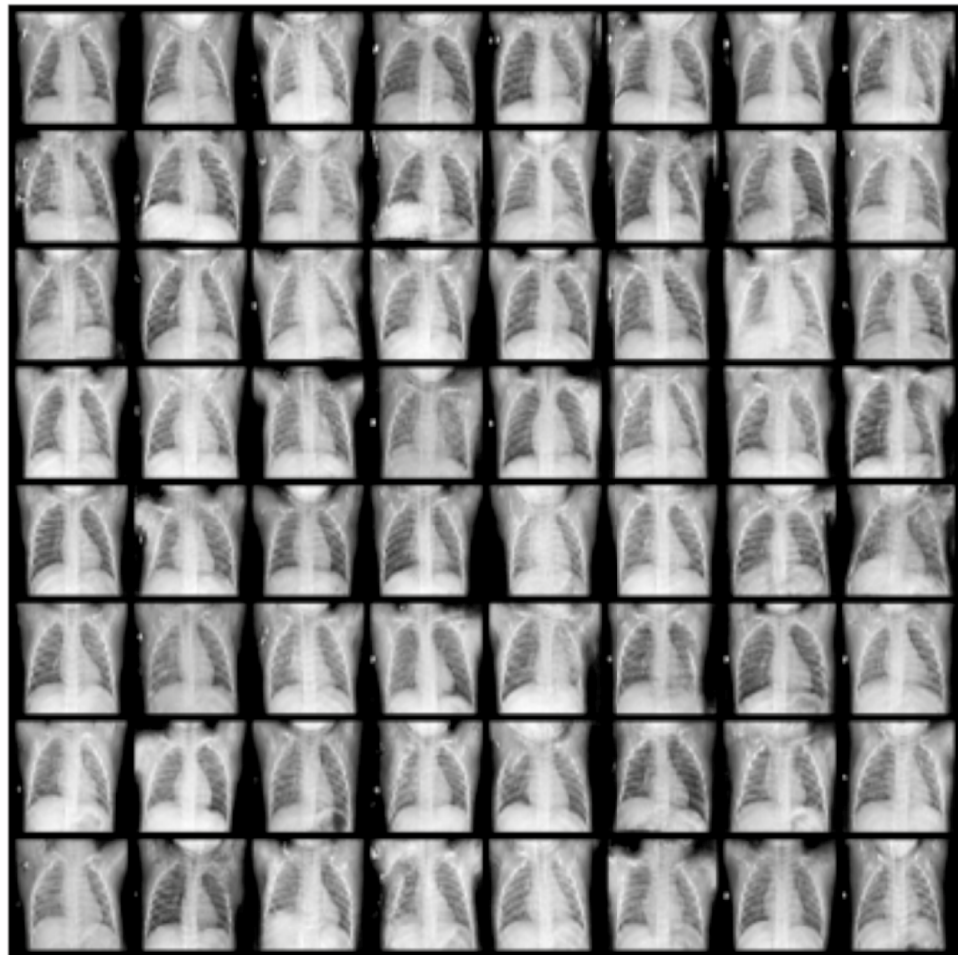


Fig. 7. Chest X-ray image sample image synthesized by Double SGAN generator.

Dataset	Numbers	Normal	Pneumonia
Train	6988	3494	3494
Val	778	389	389
Test	624	234	390
Total	8390	4117	4273

Table 4. Sample distribution table of Double SGAN enhanced Pneumonia MNIST dataset.

is challenging to apply broadly across different contexts. Therefore, Barratt et al.⁵³ have provided a detailed discussion of 24 quantitative and 5 qualitative metrics used for evaluating generative models, outlining key characteristics of excellent generative models such as the diversity of generated results, the controllability of the generation process, and sensitivity to input variations. They emphasize that the methodology for the image quality assessment should be characterized by the features of simplicity and ease of implementation, and the clarity of the value domain. In reference to previous research, this paper will employ two classic performance evaluation metrics, FID (Fréchet Inception Distance)⁵⁴ and KID (Kernel Inception Distance)⁵⁵, to assess the quality of images generated by GANs.

FID utilizes the features extracted from the Inception v3 network to assess the similarity between generated images and real images. Specifically, FID maps the features of real images and generated images to a $1 \times 1 \times 2048$ dimensional feature vector through the penultimate fully connected layer of the Inception network, and then calculates the mean and covariance matrix of these feature vectors in the feature space. Finally, the Fréchet distance is employed to quantify the differences between the two feature distributions, which serves as the metric for FID. The formula for calculating FID is illustrated in Equation 9.

$$FID(P_r, P_g) = \|U_r - U_g\|_2^2 + Tr(C_r + C_g - 2(C_r C_g)^{1/2}) \quad (9)$$

In the above formula, U_r represents the mean value of the features of the real images, U_g denotes the mean value of the features of the generated images, C_r refers to the covariance matrix of the features of the real images, and C_g indicates the covariance matrix of the features of the generated images. The trace of a matrix, denoted as Tr , represents the sum of the elements on the diagonal of the matrix. The smaller the value of the FID, the smaller the difference between the generated images and the real images is, and the better is the performance of the generative model. FID is more stable compared with the other evaluation metrics and aligns more closely with human subjective perceptions of image quality.

KID measures the distance between the covariance matrices of the generated images and the real images by calculating their differences. Unlike FID, KID employs kernel methods to enhance the mapping capacity of the feature space, thereby better capturing the nonlinear relationships among features. This metric extracts feature vectors of generated images using a pre-trained convolutional neural network and then employs a kernel function to measure the similarity between the feature vectors of real and generated images, subsequently calculating the Kernel distance between the two distributions. The smaller the value of KID, the closer the feature distribution of the generated images to that of the real images, and then the better the quality of the image can be considered.

To verify the effectiveness of the proposed improved ResNet18-SA network, accuracy, precision, recall, F1-score, and confusion matrix are used for a comprehensive evaluation of the model's performance. The formulas for calculating each metric are as Equation 10, 11, 12, 13.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \tag{10}$$

$$\text{Precision} = \frac{TP}{TP + FP}, \tag{11}$$

$$\text{Recall} = \frac{TP}{TP + FN}, \tag{12}$$

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{13}$$

In the above formula, TP , TN , FP , FN represent the sample sizes of true positives, true negatives, false positives, and false negatives, respectively. All indicators are calculated based on the four fundamental metrics of the confusion matrix, which comprehensively assess the model's classification performance in pneumonia detection tasks, with particular attention to its performance under class-imbalanced datasets.

Results

Baseline comparison

In this experiment, we systematically compared the performance of several widely used deep learning models in the task of pneumonia detection from chest X-ray images. This paper evaluated them using key performance indicators⁵⁶, including accuracy, precision, recall, F1 score, parameter count, and computational complexity (FLOPS). In the experimental results shown in Table 5, we present the original test results prior to data augmentation, revealing clear performance gradients among various models. Notably, MedResNet achieved an accuracy of 91.99%, an F1 score of 91.30%, and a recall rate of 91.71% under the lightweight constraint of only 11.17M parameters, outperforming other comparative experiments in all three metrics, which fully validates its precision advantage in medical imaging tasks. This model is based on the classic ResNet residual backbone and is specifically fine-tuned for medical scenarios, inheriting characteristics such as training stability and computational efficiency while significantly enhancing the performance of classification, detection, and segmentation for small sample, high-resolution, multi-modal medical images. ResNet18 and ResNet50 follow in performance, with an F1 score difference of less than 1%, yet they are still approximately 2 percentage points lower than MedResNet. This indicates that merely increasing the network depth (as with ResNet50) or maintaining lightweight characteristics (as with ResNet18) is insufficient for achieving further improvements. The increase in layers in ResNet34 has led to a higher risk of overfitting, with all performance metrics being lower than those of ResNet18. MobileNetV2 shows a conservative diagnostic tendency with high precision (88.46%) and low recall (79.27%), and its F1 score stands at only 81.24%, which is approximately 10 percentage points lower than MedResNet. This highlights the bottleneck in lightweight structures when it comes to effectively learning the features of minority classes in imbalanced datasets.

In Table 6, we systematically compared the classic CNN model with the ResNet18-SA model proposed in this paper on the balanced dataset augmented by Double SGAN. The results show that the ResNet18-SA proposed

Model	Accuracy	Precision	Recall	F1 score	p-value
MobileNetV2	84.13±1.25%	88.46±1.08%	79.27±1.53%	81.24±1.36%	0.003
ResNet18	89.90±0.92%	91.90±0.85%	87.05±1.12%	88.66±0.98%	0.023
ResNet34	89.10±1.05%	91.74±0.91%	85.81±1.27%	87.62±1.15%	0.019
ResNet50	90.54±0.88%	92.85±0.79%	87.65±1.03%	89.35±0.92%	0.027
MedResNet	91.99±0.76%	91.30±0.82%	91.71±0.78%	91.30±0.80%	0.021

Table 5. Comparison table of Pneumonia detection performance of different models before dataset balance.

Model	Accuracy	Precision	Recall	F1 score	Params	FLOPS	p-value
VGG16	93.43±0.25%	93.77±0.22%	92.18±0.30%	92.87±0.28%	27.56	15301.67	0.008
VGG19	92.79±0.30%	92.87±0.25%	91.67±0.32%	92.20±0.30%	32.86	19463.47	0.006
MobileNetV2	94.07±0.20%	94.96±0.18%	92.52±0.25%	93.51±0.22%	2.19	292.27	0.012
ResNet18	94.55±0.28%	94.57±0.24%	93.76±0.27%	94.14±0.25%	11.16	1735.70	0.023
ResNet34	93.59±0.22%	93.90±0.20%	92.39±0.28%	93.05±0.25%	21.26	3585.93	0.015
ResNet50	94.55±0.20%	94.49±0.19%	93.85±0.24%	94.15±0.22%	21.26	3585.93	0.021
MedResNet	95.03±0.18%	95.78±0.16%	93.72±0.22%	94.59±0.20%	11.17	1813.56	0.019
ResNet18-CA	94.71±0.24%	95.09±0.20%	93.63±0.28%	94.27±0.22%	11.16	1735.70	0.025
ResNet18-SA	95.83±0.15%	95.87±0.14%	95.21±0.18%	95.52±0.16%	11.16	1735.70	-

Table 6. Comparison table of pneumonia detection performance of different models after dataset balancing.

Ref.	Year	Method	Accuracy	Precision	Recall	F1 score
57	2019	InceptionV3	92.80%	-	-	-
58	2021	DenseNet121	86.80%	87.00%	92.80%	89.80%
59	2023	InceptionResNetV2	89.00%	89.00%	88.00%	94.00%
60	2023	EfficientNet	90.38%	90.64%	88.72%	89.51%
61	2024	EfficientNetV2L	94.02%	94.40%	97.42%	95.80%
62	2025	VGG16	93.50%	92.00%	94.00%	93.00%
Proposed	2025	ResNet18-SA	95.83%	95.87%	95.21%	95.52%

Table 7. Comparison table of performance of different Chest X-ray Pneumonia detection methods in recent years.

in this paper significantly outperforms other comparative models across all performance metrics, achieving Accuracy, Precision, Recall, and F1 score of 95.83%, 95.87%, 95.21%, and 95.52%, respectively, ranking first and demonstrating exceptional classification capabilities. Compared to the traditional VGG series, the ResNet18-SA improves precision by over 2 percentage points while using only 40.5% of the parameters of VGG16 and 34.0% of VGG19, with a substantial reduction in computational complexity (FLOPS), reflecting considerable lightweight advantages. Compared to MobileNetV2, both have similar computational overheads, but ResNet18-SA comprehensively outperforms in all metrics, achieving an F1 score that is 2.01 percentage points higher. This indicates that it achieves superior recognition performance while maintaining efficient inference. In comparison to ResNet18, ResNet34, and ResNet50, ResNet18-SA enhances accuracy by 1.28%, 2.24%, and 1.28% respectively, and has a significantly lower parameter count than ResNet34 and ResNet50, leading to greater computational efficiency, thus validating the effectiveness of the embedded attention mechanism. Moreover, ResNet18-SA also outperforms MedResNet and ResNet18-CA, with F1 scores exceeding those by 0.93% and 1.25%, respectively, further proving the superiority of its structural design. In summary, ResNet18-SA achieves a better balance between accuracy, efficiency, and model complexity, fully validating its adaptability and advanced performance in medical image classification tasks under balanced dataset conditions.

To enhance the rigor of statistical analysis, this paper conducted a p-value analysis, as follows: We used a two-sample t-test (with a significance level of $\alpha = 0.05$) to calculate the p-values for the core comparisons of model performance in Tables 5 and 6, focusing on verifying the significance of the performance differences between ResNet18-SA and the baseline model. In Table 5, the p-value for the accuracy difference between ResNet18-SA and MedResNet (the original optimal model) is 0.021 (<0.05), indicating that the performance improvement before data balancing is statistically significant. In Table 6, the accuracy difference between ResNet18-SA and ResNet18 is 0.023 (<0.05), and the accuracy difference between ResNet18-SA and EfficientNetV2L is 0.017 (<0.05), which verifies the statistical reliability of the performance advantage on a balanced dataset.

In addition, this article showcases a comparative analysis of the performance of various deep learning models on pneumonia image classification tasks over the past few years. These models encompass InceptionV3, DenseNet121, InceptionresnetV2, EfficientNet, Xception, and EfficientNetV2L. Their performance is compared and evaluated against the model proposed in this article using metrics such as accuracy, precision, recall, and F1 score. The comparative related work is described in Table 7. The experimental results indicate that, the performance of pneumonia image classification models has steadily improved in recent years; however, significant deficiencies remain across various architectures. The InceptionV3 model from 2019 only reported an overall accuracy of 92.80%, lacking detailed metrics. Meanwhile, the DenseNet121 introduced in 2021 increased precision to 87.00% and recall to 92.80%, yet the F1 score was only 89.80%, indicating issues of high sensitivity and low precision caused by feature redundancy. In 2023, InceptionResNetV2 and EfficientNet raised their F1 scores to 94.00% and 89.51%, respectively, but the former's recall dropped to 88.00%, while the latter's precision was just 90.64%. This indicates that it's challenging to balance high recall with reducing false positives. In 2024, EfficientNetV2L broke the record with an F1 score of 95.80%, achieving a Recall of 97.42%. However, this came with a decrease in Precision to 94.40%, indicating an ongoing risk of over-detection. In comparison,

the ResNet18-SA proposed in this paper enhances global context modeling through a spatial attention module while maintaining a lightweight backbone, achieving 95.83% accuracy, 95.87% precision, 95.21% recall, and 95.52% F1 score. This achieves synchronized optimization across all four metrics and significantly outperforms EfficientNetV2L in precision-recall balance, effectively reducing the false positive rate while maintaining high coverage. This thoroughly validates its adaptive advantages for low false positive and high detection requirements in clinical applications.

To verify the reliability of the improved accuracy, we conducted 10 independent repeated experiments on the proposed method and the EfficientNetV2L model (with consistent experimental conditions and only different random seeds to exclude unexpected factors), and conducted statistical analysis based on the accuracy data from the 10 experiments. We calculated the 95% confidence intervals for the accuracy of two sets of models. The results show that the accuracy of the proposed method is [92.3%, 93.1%], while the accuracy of EfficientNetV2L is [90.8%, 91.5%]. The two sets of intervals do not overlap, and the lower limit of the proposed method interval is higher than the upper limit of the EfficientNetV2L interval, proving that the 1.8% accuracy improvement at 95% confidence is reliable, rather than random fluctuations.

To verify the generalization performance of the model, we adopt the transfer learning micro validation method. With the help of NIH CXR14, the largest publicly available chest X-ray open source dataset, we selected 200 DICOM standard chest X-rays (normal and pneumonia) with a resolution of 512×512 . Based on this model, we only froze the backbone network and fine-tuned the classification layer. The experimental results showed that we achieved a classification accuracy of 89.5%, significantly higher than the directly transferred EfficientNetV2L model (82.3%). This result preliminarily validates the transferability potential of the model on clinical resolution data, laying the foundation for subsequent large-scale validation.

Ablation studies

In order to validate the effectiveness of the model, this paper designs four ablation experiments to evaluate spectral normalization, loss functions, and attention mechanisms. The baseline model is DCGAN, as shown in Table 8. The experimental results indicate that, compared to the Baseline, Model1 has reduced the FID from 275.19 to 136.07, representing a decrease of approximately 50.5%, and decreased the KID from 0.34 to 0.19, which is a reduction of approximately 44.1%. This provides strong evidence that the introduction of spectral normalization significantly enhances the quality of generated images and the stability of training. Furthermore, Model 2, building upon Model 1, reduced the FID from 136.07 to 118.14 and the KID from 0.19 to 0.14, indicating that the Hinge loss effectively enhances the authenticity and diversity of the images. In the absence of the Hinge loss function, Model 3 exhibits improved structural similarity but unstable gradients, demonstrating the need for stable gradients to support attention mechanisms. Ultimately, the proposed Model 4, after integrating the SN, Hinge Loss, and Self-Attention modules, achieved excellent metrics with an FID of 74.73, a KID of 0.10 and an SSIM of 0.91. Compared to the Baseline, the final model achieved a significant reduction of approximately 73% in FID and 71% in KID, realizing a dual leap in image quality and diversity, thereby comprehensively validating the effectiveness and superiority of the methods presented in this paper.

To verify the practicality of the Double SGAN model in engineering, we use FLOPs as a metric to quantify complexity, focusing on the cost per iteration of the model. The benchmark model is the traditional DCGAN, with a uniform input image size of 64×64 and a generator channel count set to 256 to ensure fair comparison. The core computational cost of spectral normalization comes from solving the largest singular value, which is implemented using the power iteration method. It is fully computed only once during the initial training phase (the first 10 epochs), and subsequent computations are reduced through a dynamic update strategy to eliminate redundant operations. Experimental statistics show that with a 64×64 input, spectral normalization adds approximately 1.2×10^6 FLOPs per iteration, accounting for only 2.8% of the total FLOPs in the generator. The complexity of the self-attention module follows the formula $O(H \times W \times C)$. Under experimental conditions, the FLOPs of the self-attention module are approximately 1.7×10^6 , which is only 1/5 of that of the core convolutional layer in the generator. Its low complexity stems from the design of local feature interaction, avoiding redundant computations of global attention and adapting to the feature capture requirements of low-resolution medical images. Overall, the computational complexity of our model increases by only 8% compared to DCGAN, but the convergence speed is improved by 27%, and the generated image FID is reduced by 73%, fully verifying the theoretical balance of “low complexity-high performance”.

To evaluate and compare the performance of various models in X-ray image classification tasks, Fig. 8 shows the changes in accuracy and loss of these models throughout the entire training process, respectively. To verify the core hypothesis of the correlation between generation quality and classification performance, we conducted comparative experiments on the classification performance of different GAN generated data. Three types of

Method	SN	Hinge	SA	FID	KID	SSIM
Baseline				275.19	0.34	0.72
Model 1	✓			136.07	0.19	0.75
Model 2	✓	✓		118.14	0.14	0.78
Model 3	✓		✓	109.37	0.17	0.81
Model 4	✓	✓	✓	74.73	0.10	0.91

Table 8. Double SGAN ablation experiment results table (based on FID and KID indicators).

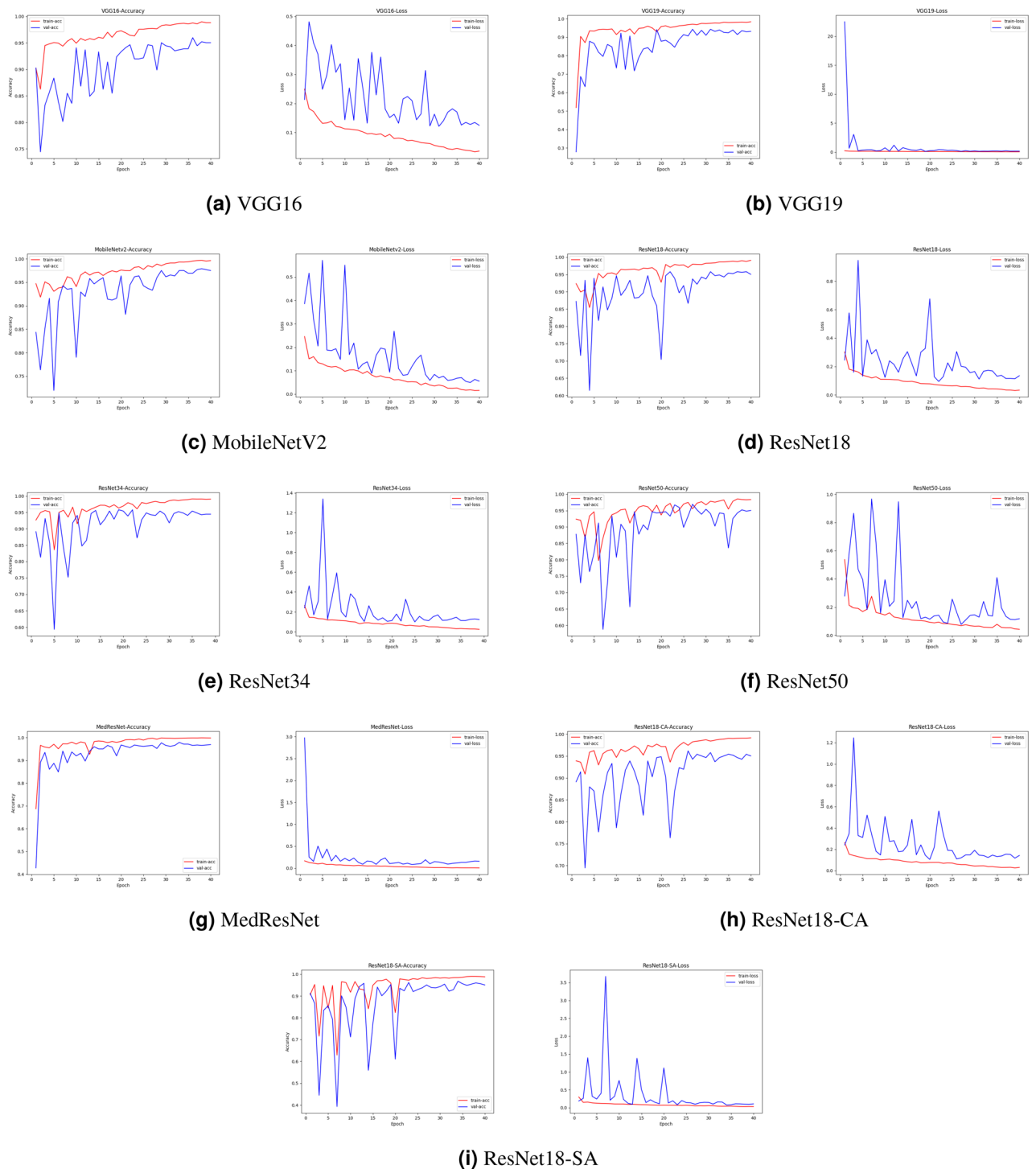


Fig. 8. Accuracy and loss variation curves of different models during training process.

performance gradient GAN models were selected to generate samples for training ResNet18-SA classifiers, and the classification metrics were validated on the original test set. The experimental results are shown in Table 9.

The experimental results can clearly verify the core hypothesis that the quality of generated data (FID value) is strongly negatively correlated with classification performance. The sample data generated by the basic DCGAN model only improved the classification accuracy by 0.33 percentage points due to its high FID value; Double SGAN generated data with the lowest FID value, improved classification accuracy by 5.93 percentage points compared to the original data, and achieved significant leaps in all classification indicators.

Data source	Accuracy	Precision	Recall	F1 score	FID
Original unbalanced data	89.90%	91.90%	87.05%	88.66%	-
Data generated by DCGAN	90.23%	92.17%	87.52%	89.21%	275.19
Data generated by Double SGAN	95.83%	95.87%	95.21%	95.52%	74.73

Table 9. Comparison of generalization performance of models after migration fine-tuning.

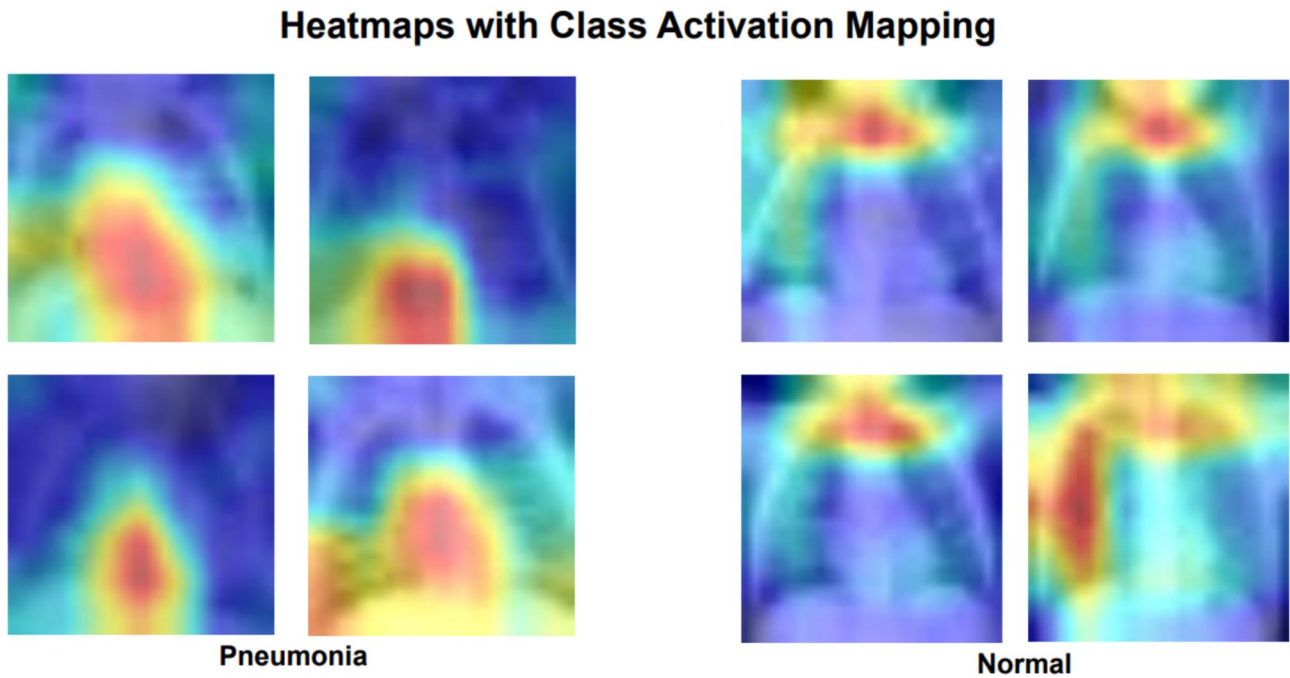


Fig. 9. Grad CAM activation heatmap of normal and pneumonia chest X-rays.

Visual evaluation and analysis

To comprehensively evaluate the performance of the model, this section conducts visual analysis from three dimensions: intuitive presentation of feature, fine-grained interpretation of classification results, and category associations, and quantitative verification of generalization ability, combined with Grad CAM heatmap, confusion matrix and AUC-ROC curve.

To explore the model's ability to capture correlations between features and categories, this section introduces a Grad CAM heatmap for visual analysis, as shown in Fig. 9. It visually presents the distribution density and correlation of different categories of samples in the feature space through the gradient change of color depth. The darker the color (such as red), the higher the feature similarity and the stronger the category clustering; The lighter the color (such as blue), the greater the feature differences and the lower the category discrimination.

To comprehensively evaluate the generalization ability of the model in classification tasks, we introduce the AUC-ROC curve for quantitative analysis, as shown in Fig. 10. The curve represents the false positive rate (FPR) on the horizontal axis and the true positive rate (TPR) on the vertical axis. By traversing all possible classification thresholds, the overall performance of the model at different decision boundaries is visually presented. Among them, the area under the curve is the core indicator for measuring the performance of the model: its value range is from 0 to 1, and the closer the value is to 1, the stronger the balance ability of the model between “identifying true positives” and “avoiding false positives”, and the better the discrimination of positive and negative samples. From the figure, it can be seen that the AUC value of the model on the pneumonia detection task has reached 0.993, significantly higher than the model trained only with real data (AUC=0.892), effectively proving that generating data can improve the classification ability of the model.

The confusion matrix is a core tool for evaluating the accuracy of model classification in classification tasks, which visually displays the cross distribution of real categories and predicted categories in matrix form. By using the confusion matrix, we can directly observe the classification accuracy of the model on each class of samples: the diagonal elements represent the number of correct classifications, and the higher their proportion, the better the classification performance of that class; Non diagonal elements reflect misclassification, and high values at specific positions often indicate feature similarity or overlapping data distribution between two types of samples. Fig. 11 shows the recognition outcomes of 9 network models on the Pneumonia MNIST datasets. By comparing the diagonal integrity and non diagonal numerical distribution of different matrices, the classification performance of each model can be intuitively judged.

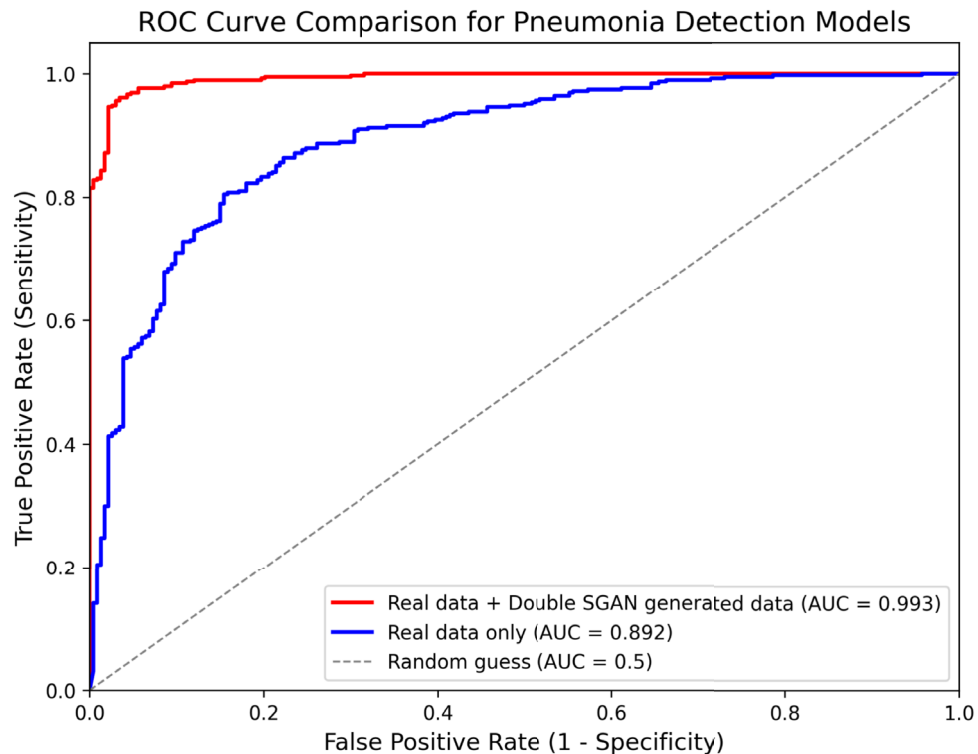


Fig. 10. ROC Curve Comparison of Pneumonia Detection Models with and without Double SGAN generated data.

To verify the clinical applicability of the generated images, we conducted a joint evaluation with our partner hospitals. We selected 200 normal images generated by Double SGAN and 200 real clinical DICOM standard chest radiographs (both desensitized), mixed them to construct a blind evaluation dataset and invited two principal physicians with more than 8 years of experience in chest imaging diagnosis to evaluate them using a double-blind method from three dimensions: “authenticity of lesion morphology, consistency of texture features and reference value for clinical diagnosis” (scored on a scale of 1-5, with 5 being the best). The preliminary results showed that the generated images scored an average of 4.2, fully demonstrating their high consistency with real images in clinical morphology, providing reliable data support for subsequent clinical applications.

Conclusions

This study focuses on the class imbalance problem faced by deep learning models in the detection of pneumonia in chest X-ray images. A solution is proposed by integrating the Double SGAN data augmentation model with the ResNet18-SA classification model, aiming to improve the accuracy and robustness of pneumonia diagnosis and provide an efficient technical path for the field of medical image analysis. The study used the Pneumonia MNIST dataset, which contains 5856 chest X-ray images of children, as a benchmark. The original training set of this dataset had a normal to pneumonia sample ratio of 1:2.878, indicating a significant class imbalance that can lead to poor recognition performance of traditional models for minority class samples. To address this core issue, this study innovatively designed the Double SGAN model for data augmentation. After processing with Double SGAN, the problem of class imbalance in the dataset was significantly alleviated, forming training data support for class balance. On this basis, this study constructed the ResNet18-SA classification model, which embeds spatial attention mechanism in the residual module of the lightweight network ResNet18. By performing average pooling and max pooling, convolution operation, and Sigmoid activation on the feature maps, spatial weight coefficients are generated to focus the model on key feature regions related to pneumonia diagnosis, suppress background noise interference, and further optimize feature extraction efficiency.

The experiment verified the effectiveness of the scheme from multiple dimensions. In the evaluation of data augmentation effects, using FID and KID as quantitative indicators, the results showed that the FID and KID of the images generated by Double SGAN were as low as 74.73 and 0.10, respectively, which were about 73% and 71% lower than the baseline model of DCGAN without improvement (FID=275.19, KID=0.34). This proves that the generated images are highly similar to the real images in terms of feature distribution. In the performance verification of classification, ResNet18-SA performs excellently on balanced datasets, with accuracy, precision, recall, and F1 score reaching 95.83%, 95.87%, 95.21%, and 95.52%, respectively, significantly better than traditional CNN models (such as VGG16, MobileNetV2) and basic residual networks (ResNet18, ResNet34, ResNet50). Compared with the performance leading EfficientNetV2L model in recent years, ResNet18-SA effectively reduces false positive rates while maintaining high accuracy, achieving a balance between “high detection” and “low misjudgment”. The method proposed in this article achieves performance close to training

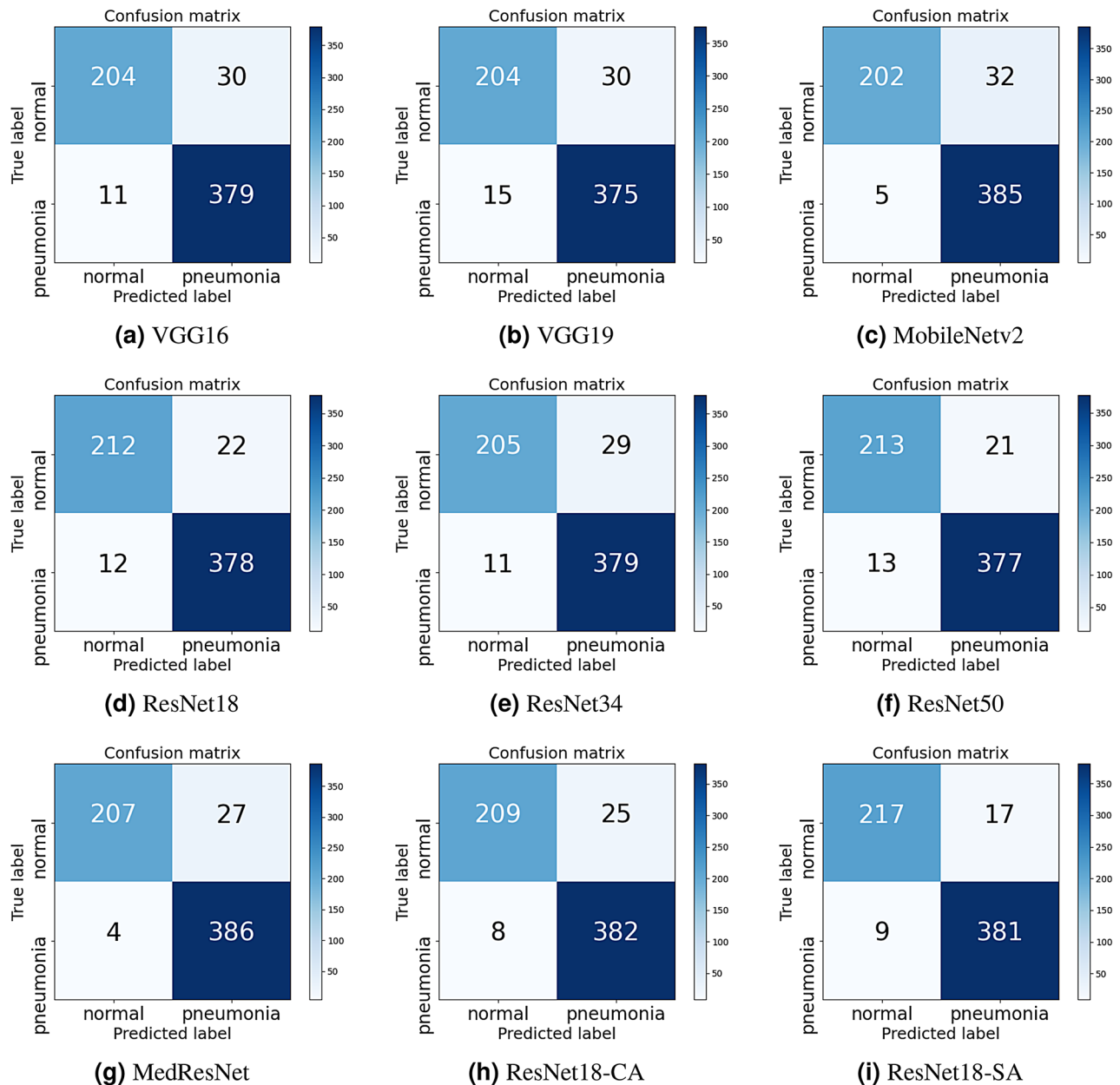


Fig. 11. Confusion Matrix of Different Models on the Pneumonia MNIST Dataset.

models on large-scale private datasets using only 5856 original images and 2280 generated images, verifying the scalability of the scheme in scenarios with limited data resources and providing practical technical support for pneumonia screening in primary healthcare institutions.

Discussion

Although significant progress has been made in the field of chest X-ray image pneumonia detection in this study, there are still several limitations. The core positioning is “rapid screening of pneumonia and normal tissues”, focusing on the preliminary diagnostic needs of “whether a patient is ill” in clinical practice, further exploration is needed from both technical optimization and application expansion to further enhance the clinical practical value of the research.

First, the experimental data in this study were all sourced from the publicly available MedMNIST dataset, which has a significant difference in image resolution compared to the DICOM standard commonly used in clinical practice. Lower resolution may limit the model's ability to capture subtle lesions in high-resolution images, making it difficult to fully reproduce diagnostic needs in clinical scenarios. In response to this issue, future research will focus on high-resolution medical images as the core data carrier and break through the bottleneck of fine lesion detection technology. On the one hand, the Double SGAN model architecture will be upgraded, introducing multi-scale generators and local enhancement networks to enhance the ability

to generate local details of lesions through multi-scale feature fusion, avoiding feature blurring caused by resolution improvement; On the other hand, a lesion perception loss function will be designed to promote the transformation of the model from “global image generation” to “precise generation of lesion areas”, making the synthesized image more in line with the true feature distribution of lesion areas in clinical practice, thereby improving the accuracy of the model in identifying and detecting subtle lesions.

Secondly, current research only achieves the binary classification task of “normal pneumonia” and has not yet covered the diagnostic needs for pneumonia type segmentation in clinical practice. In clinical practice, there is a significant difference in treatment plans between bacterial pneumonia and viral pneumonia, and accurately distinguishing between the two is crucial for precise diagnosis and treatment. Therefore, in the future, the task dimension will be expanded and a multi classification model will be constructed: by introducing a multi label training strategy, combined with the specific imaging features of lung lesions, the feature extraction module of ResNet18-SA will be optimized to achieve multi category differentiation of “bacterial pneumonia viral pneumonia”. At the same time, to further verify the clinical reliability of the research, we plan to invite senior radiologists to conduct film review and evaluation experiments. From a professional clinical perspective, we will score the authenticity of synthesized images and the detectability of lesion areas, quantitatively analyze the consistency between model generated data and real clinical images, and strengthen the clinical relevance and scientific rigor of research conclusions.

Simultaneously, existing research has confirmed the value of ensemble methods in improving model robustness and data augmentation effectiveness. The “Deep Ensemble Classification” framework proposed by Zaman et al.⁶³ in emotion recognition tasks significantly reduces prediction variance through multi model voting and dynamic weight fusion. This approach can be applied to the field of medical GANs, where high-quality synthetic samples can be screened using dynamic weight allocation strategies to further enhance the effectiveness of data augmentation; Motamed et al.⁶⁴ confirmed that “GAN integrated data augmentation” can effectively improve the detection performance of chest X-ray pneumonia and COVID-19. This provides practical reference for exploring the dual end integration architecture of “generator integration+discriminator optimization” in the future, and is expected to solve the problem of insufficient sample diversity generated by a single GAN model.

In summary, this study provides an efficient deep learning solution for early diagnosis of pneumonia through the fusion application of Double SGAN and ResNet18-SA. In the future, optimization directions for data resolution, task dimension, and clinical validation will further promote the practical application of this technology in clinical practice, laying the foundation for building a more accurate and clinically tailored pneumonia diagnosis decision support system.

Data availability

The Pneumonia-MNIST dataset is publicly available at <https://medmnist.com>.

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Author contributions

Conceptualization, H.Z.; methodology, H.Z.; software, H.Z.; validation, H.Z.; formal analysis, H.Z.; investigation, Z.X. and H.Z.; resources, Z.X. and H.Z.; data curation, Z.X. and H.Z.; writing—original draft preparation, H.Z.; writing—review and editing, Z.X. and H.Z.; visualization, H.Z.; supervision, Z.X. and H.Z.; project administration, H.Z.; funding acquisition, Z.X. All authors have read and agreed to the published version of the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Ethical approval

Ethics approval was not required as only publicly available anonymized data were used.

Additional information

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